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Article in *Journal of Medical Microbiology* · October 2020

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Molecular evaluation of quorum quenching potential of vanillic acid against *Yersinia enterocolitica* through transcriptomic and *in silico* analysis

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Abstract

Introduction. *Yersinia enterocolitica* is one of the leading food-borne entero-pathogens causing various illnesses ranging from gastroenteritis to systemic infections. Quorum sensing (QS) is one of the prime mechanisms that control the virulence in *Y. enterocolitica*.

Hypothesis/Gap Statement. Vanillic acid inhibits the quorum sensing and other virulence factors related to *Y. enterocolitica*. It has been evaluated by transcriptomic and *In silico* analysis. Therefore, it can be a prospective agent to develop a therapeutic combination against *Y. enterocolitica*.

Aim. The present study is focused on screening natural anti-quorum-sensing agents against *Y. enterocolitica*. The effect of selected active principle on various virulence factors was evaluated.

Methodology. In total, 12 phytochemicals were screened by swarming assay. MATH assay, EPS and surfactant production assay, SEM analysis, antibiotic and blood sensitivity assay were performed to demonstrate the anti-virulence activity. Further, RNA sequencing and molecular docking studies were carried out to substantiate the anti-QS activity.

Results. Vanillic acid (VA) has exhibited significant motility inhibition, thus indicating the anti-QS activity with MQIC of 400 µg ml⁻¹ without altering the cell viability. It has also inhibited the violacein production in *Chromobacterium violaceum* ATCC 12472, which further confirms the anti-QS activity. VA has inhibited 16% of cell-surface hydrophobicity (CSH), 52% of EPS production and 60% of surfactant production. Moreover, it has increased the sensitivity of *Y. enterocolitica* towards antibiotics. It has also made the cells upto 91% more vulnerable towards human immune cells. The transcriptomic analysis by RNA sequencing revealed the down regulation of genes related to motility, virulence, chemotaxis, siderophores and drug resistance. VA treatment has also positively regulated the expression of several stress response genes. In furtherance, the anti-QS potential of VA has been validated with QS regulatory protein YenR by *in silico* molecular simulation and docking study.

Conclusion. The present study is possibly the first attempt to demonstrate the anti-QS and anti-pathogenic potential of VA against *Y. enterocolitica* by transcriptomic and *in silico* analysis. It also deciphers that VA can be a promising lead to develop biopreservative and therapeutic regimens to treat *Y. enterocolitica* infections.

INTRODUCTION

Yersinia enterocolitica is a Gram-negative gastrointestinal bacterial pathogen that is distributed worldwide. It is the

most common *Yersinia* species that infects humans and animals. Generally it spreads through contaminated food or water. It has the potential to survive out of the mammalian host and can grow even in 4–42 °C [1]. *Y. enterocolitica* has

Received 27 May 2020; Accepted 25 September 2020; Published 21 October 2020

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Keywords: Quorum sensing; *Yersinia enterocolitica*; Vanillic acid; Transcriptomic analysis; Molecular docking.

Abbreviations: CSH, cell surface hydrophobicity; EPS, extra polymeric substances; MATH, microbial adherence to hydrocarbons; MQIC, minimal quorum inhibitory concentration; QS, quorum sensing; SEM, scanning electron microscope; VA, vanillic acid.

†These authors contributed equally to this work

Three supplementary videos, nine supplementary figures and three supplementary tables are available with the online version of this article.

been isolated in various food substances such as pork [2], beef [3], chicken meat [4], sea foods [5], vegetables [6] and dairy products [7]. It causes Yersiniosis ranging from self-limiting gastroenteritis with abdominal pain and diarrhoea to life-threatening bacteraemia, rarely. The rate of infection is high in children and infants, in whom the gastroenteritis can develop as fatal systemic infection with 50% mortality [8]. It plays a vital role in the pathology of appendicitis [9]. Food-associated outbreaks of *Y. enterocolitica* have been documented in USA, UK, Norway, Sweden, Hungary, Finland, Germany, Australia, Korea, Bangladesh and India [10–14]. Earlier studies have stated that *Y. enterocolitica* is susceptible to a major group of antibiotics and is rarely prescribed for mild infection, nonetheless for systemic diseases and bacteraemia, strong antibiotic treatment is required [15]. In recent years, the prevalence of multidrug resistance in *Y. enterocolitica* is escalating rapidly worldwide, which becomes a hectic obstacle for the health sector to treat patients [16–18].

Quorum sensing (QS) is a communication mechanism between cells via chemical signals, which contributes to the establishment of bacterial community in its niche by synchronized genetic expression of virulence. The QS system of *Yersinia* works on LuxI/LuxR orthologues called *yenI/yenR* that employs AHLs as signals [19, 20]. An AHL-mediated QS system operates swarming motility, a major virulence factor by controlling the expression of flagellar genes such as *flhDC*, *fliA*, *fleB* [19, 21]. Moreover, QS regulates and determines the expression of virulence and indirectly influences the drug resistance in many bacteria [22].

Phytochemicals are widely used as promising bioactives for various diseases since time immemorial [23]. As they have a huge chemical diversity, an anti-QS active principle against *Y. enterocolitica* is highly envisaged. They are also known to be non-toxic, cost effective and agents with a high degree of applicability in food system. A natural anti-QS agent against *Y. enterocolitica* will inhibit the virulence and sustainability in the host and thereby it can be managed by combinatorial antibiotic therapy. With this background, in the search for natural therapeutic agents that ameliorate the clinical condition in Yersiniosis, phytochemicals were chosen to screen for anti-QS agents against *Y. enterocolitica*. The potential of selected phytochemicals to inhibit QS-based swarming and other virulence factors such as EPS production, cell-surface hydrophobicity, surfactant production has been investigated. In furtherance, the effect of the bioactive lead on sensitivity of the pathogen with antibiotics and human blood was analysed.

METHODS

Bacterial strain and culture condition

The bacterial strain of *Y. enterocolitica* (MTCC 3235) was isolated from beef. Routine growth and maintenance of *Y. enterocolitica* strain was done on Luria–Bertani agar (LBA) plates at 26 and 4°C, respectively. They were also preserved as glycerol stocks at –80°C. Similarly, *Chromobacterium violaceum* ATCC 12472 was subcultured and maintained in LB agar plates.

Liquid cultures were made by picking up an isolated colony, inoculated in LB broth and grown at 26°C for 24h.

Phytochemicals and antibiotics stocks

Phytochemicals of various groups such as flavonones, flavonoids, flavones, polyphenolic compounds, terpenoids, peptides, alkaloids and fatty acids were used for this study and they were acquired from Alfa Aesar (Hyderabad, India). Thymol, squalene, 4,7-dihydroxy isoflavone, caffeine, vanillic acid, geraniol, 5-dodecanolide, oleic acid, α -methyl benzyl amine, 4-phenyl butyric acid and glutathione were dissolved in 70% ethanol while hesperidine was dissolved in DMSO. All phytochemicals were generally recognized as safe (GRAS) grade. The stock solutions of phytochemicals were made at 10 mg ml⁻¹ concentration. The antibiotic discs of gentamycin (30 µg), amikacin (30 µg), ampicillin (10 µg), azithromycin (15 µg), ciprofloxacin (5 µg), chloramphenicol (30 µg) were procured from HiMedia (India). The powder form of these antibiotics were also procured (HiMedia, India) and stocks were made at 10 mg ml⁻¹.

Swarming assay

Swarming motility assay was carried out as described by [24, 25] with considerable modifications. The detailed procedure is given in the Supplementary Material (available in the online version of this article).

Antimicrobial susceptibility test

Antibiotic susceptibility was checked by antibiotic disc diffusion assay as per CLSI standard protocol M02-A11 [26] with suitable modifications. The detailed procedure is given in the Supplementary Material.

Broth microdilution assay

Broth microdilution assay was performed as done by [27]. The method was described in detail in the Supplementary Material.

Quantification of extra polymeric substances

EPS assay was carried out as described by [28]. The detailed procedure is given in the Supplementary Material.

Scanning electron microscopic (SEM) analysis

Control and treated cells were prepared for SEM analysis and the specimens were observed under SEM (Hitachi, Model: S-3400N) with magnifications of 6000, 10000 and 20000× [24, 29]. The detailed procedure is given in the Supplementary Material.

Microbial adherence to hydrocarbons (MATH)

The MATH assay was performed to measure the change in cell-surface hydrophobicity (CSH) upon VA treatment on of *Y. enterocolitica* as described by [30]. The detailed procedure is given in the Supplementary Material.

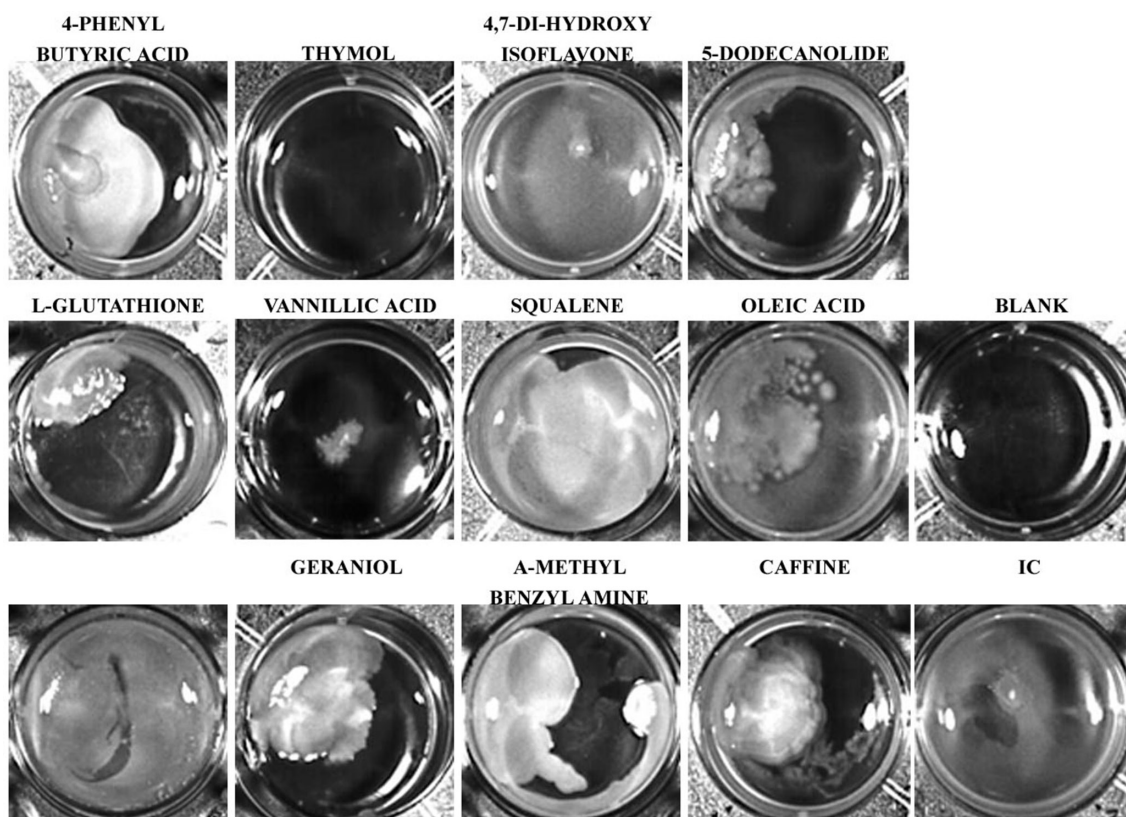


Fig. 1. Swarming assay of *Y. enterocolitica* to screen phytochemicals at 500 µg ml⁻¹.

Foaming ability assay

This assay was performed to analyse the surfactant production between VA treated and untreated *Y. enterocolitica* comparatively [31].

Blood sensitivity assay

Cell suspensions of *Y. enterocolitica* with and without VA were prepared and tested for blood sensitivity as described by [27, 32]. The experiment was done with an appropriate ethical clearance from Human ethical committee, Pondicherry University (No: PU/FST/HEC/2018-19/17) [27, 32].

Quantification of violacein

The violacein inhibition assay was performed to corroborate the anti-QS ability of VA using *C. violaceum* ATCC 12472 culture [30]. The detailed procedure is given in the Supplementary Material.

Whole transcriptomic analysis

Total RNA was isolated from control and treated bacterial cells followed by quality and quantity assessment. The selected RNA samples were proceeded for library preparation and sequencing analysis as described by [33, 34]. The detailed procedure is given in the Supplementary Material.

In silico analysis of anti-QS activity of VA

YenR (UniProtKB - A0A0E1NFA0) is a QS transcriptional activator of *Y. enterocolitica*. The structure of YenR was modelled by multi-template homology modelling. After validation the predicted structure was used for molecular docking with default ligand (OHL) and VA. The stability and binding strength of ligands and protein complexes were analysed by molecular dynamics analysis [35–48]. The detailed procedure is given in the Supplementary Material.

Statistics

Independent assays were carried out for all experiments with three replicates. The given values are mean values with SD. The one-way ANOVA statistical test was performed to analyse the significance using a SPSS package (SPSS v17.0; SPSS, USA). The *P* value ≤ 0.05 was set as significant.

RESULTS AND DISCUSSION

Yersiniosis has become the serious food-borne disease next to campylobacter and salmonella infection in developing countries of temperate zone [14, 49]. *Y. enterocolitica* has been isolated from a broad range of food substances such as vegetables, meat, sea foods, poultry, egg products, dairy products etc. and it infects through raw or improperly cooked

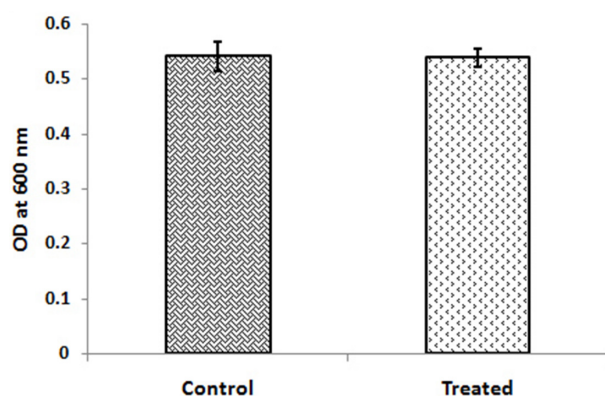


Fig. 2. Effect of vanillic acid on growth of *Y. enterocolitica*.

food stuffs [50, 51]. QS is the key factor that organizes the virulence expression in *Y. enterocolitica* [1]. In the recent era, QS has been considered as an ideal target to arrest the virulence and pathogenesis to combat microbial infections and this concept is developing and concretized by several upcoming publications and patents [52].

Screening of the anti-QS agent against *Y. enterocolitica*

Swarming motility is a perceptible virulence phenotype of *Y. enterocolitica* controlled by QS [19]. Hence in the present study, anti-QS agents were screened by swarming assay. Upon screening 12 phytochemicals, thymol was found to possess antibacterial activity and VA with a strong motility inhibition. The results showed that VA has anti-QS potential, which was reflected well in swarming inhibition (Fig. 1).

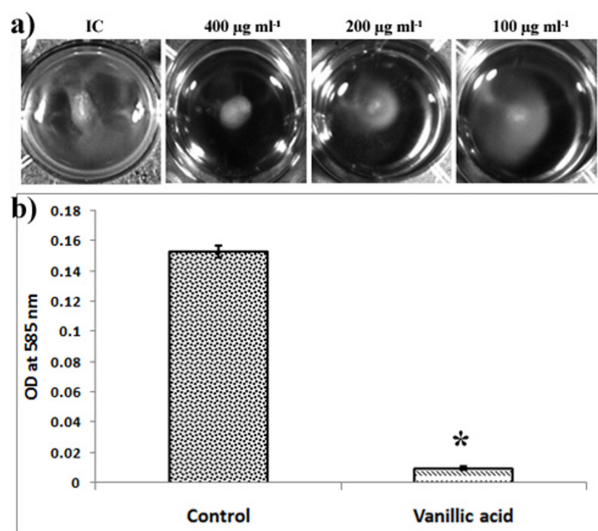


Fig. 3. (a) Effect of vanillic acid on swarming motility of *Y. enterocolitica* at different concentrations. (b) Effect of vanillic acid on violacein production in *C. violaceum*. Error bars and asterisks indicate the SD and statistical significance ($P \leq 0.05$), respectively.

Effect of VA on growth of *Y. enterocolitica*

An ideal anti-QS agent is envisaged to be a non-antimicrobial to evade Darwinian positive selection or drug resistance [24]. Hence, the present study is focused on non-antimicrobial, anti-QS agent, therefore the effect of VA on growth of *Y. enterocolitica* was tested by broth micro-dilution assay. The cultures with and without treatment were compared spectrometrically. As can be seen from the result (Fig. 2), no significant change in the growth was observed in VA treatment ($500 \mu\text{g ml}^{-1}$). The results clearly exemplified that VA has restrained the QS but not the viability and thus ensured that there will not be any eventual resistance against VA by *Y. enterocolitica*. Thus, VA was selected to proceed further. This non-antimicrobial, anti-QS activity is comparable with various anti-QS reports [24, 27–31].

Determination of MQIC

The minimal concentration that possesses maximum possible activity has been determined as MQIC through swarming assay. The VA has been tested at concentrations ranging from 100 to $400 \mu\text{g ml}^{-1}$. At $100 \mu\text{g ml}^{-1}$ no considerable change was observed, but in $200 \mu\text{g ml}^{-1}$ the swarming was substantially retarded. In the case of $400 \mu\text{g ml}^{-1}$, the migration was ceased completely and beyond which no inhibition was possible. The result indicated that swarming motility has been inhibited by VA in a dose-dependant manner (Fig. 3a). Thus the MQIC of VA was determined as $400 \mu\text{g ml}^{-1}$ at which a maximum inhibition was observed and further experiments were done at $400 \mu\text{g ml}^{-1}$.

Anti-QS potential of VA against *C. violaceum*

C. violaceum ATCC 12472 is a reporter strain for QS that produces violacein, a cell-bound purple pigment coordinated by CviI/R QS system. The anti-QS potential of VA at MQIC was corroborated by determining the violacein inhibition in *C. violaceum*. It was found that VA has reduced the violacein synthesis upto 93% (Fig. 3b). Previous reports also suggested that violacein inhibition of QS reporter strain may reflect the QS inhibition in pathogens [3, 46, 53].

Evaluation of anti-QS activity by swarming assay

The quorum quenching potential of VA has been confirmed by swarming assay. Initially during screening, the swarming assay was done in 12-well polystyrene plates, where the surface radius is less. Hence, the swarming inhibitory potential has been further confirmed in petri plates containing soft agar, through which the relative swarming ability was clearly distinguished (Fig. S1). The result evidently proves that VA has drastically inhibited the QS-regulated swarming motility.

The foresaid results authenticate the anti-QS potential of VA against *Y. enterocolitica*. These results correlate with the report of Gopu et al. [53], who demonstrated the swarming inhibition by quercetin at $80 \mu\text{g ml}^{-1}$. However, they have reported the antibacterial activity at $95 \mu\text{g ml}^{-1}$ and its synergistic interaction with antibiotics. But the present study is focused on a non-antimicrobial, anti-QS agent. Truchado et al. [54]

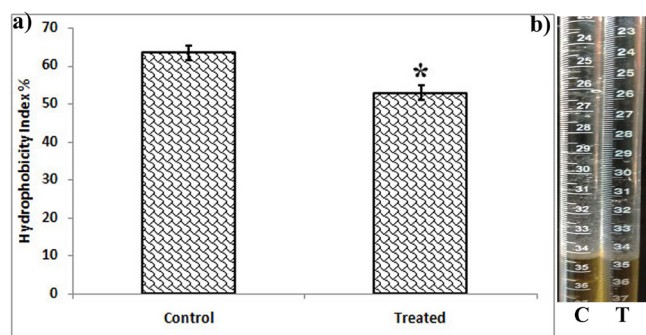


Fig. 4. (a) Effect of vanillic acid on CSH of *Y. enterocolitica*. Error bars and asterisks indicate the SD and statistical significance ($P \leq 0.05$), respectively. (b) Effect of vanillic acid on surfactant production of *Y. enterocolitica*.

have reported anti-QS activity of orange extract and its major flavanones such as naringin, neohesperidin and hesperidin without antibacterial activity against *Y. enterocolitica*. Similarly, urolithins have been reported with anti-QS activity without antibacterial activity [21]. VA is an oxidized form of vanillin, both are used as flavouring agents in food industries. It is present at maximum level in the root of *Angelica sinensis* commonly known as female ginseng and acai fruit oil '*Euterpe oleracea*'. It is also present in grapes, olives and berries. It is a well-known non-toxic phytochemical, which has already been reported with several biological activities such as antibiofilm [55], anticancer [56] and anti-inflammatory activity that cures neuro-degeneration and asthma [57, 58].

Inhibition of CSH by VA

CSH is the determining factor for cell adhesion and pathogenesis [59]. CSH has been linked with its auto-agglutination and binding with extracellular matrix proteins such as collagen, fibronectin and laminin. The removal of CSH affects the virulence in *in vivo* condition due to reduced binding with extracellular matrix proteins [60]. Lachica and Zink [61]

have reported that CSH of *Y. enterocolitica* is associated with a plasmid and it provokes the virulence. Paerregaard *et al.* [62] conjectured that the reduced adhesion of *Y. enterocolitica* to ileal brush border membrane vesicles in a rabbit model was due to reduced CSH. Hence, there is an immense need for an agent with effective anti-hydrophobicity properties, to combat pathogens by hindering contacts between pathogenic cells and the host surface. In MATH assay, control cells were predominantly shifted to organic phase due to the CSH, whereas in treated, the rate of shift was reduced. As can be seen from the result, the CSH has been reduced upto 16% in treated cells (Fig. 4a). Thus, VA can act as an anti-hydrophobicity agent, which is envisaged to reduce cell adhesion, invasion and pathogenesis of *Y. enterocolitica*.

Inhibition of EPS production by VA

Similar to CSH, EPS production is a related virulence factor that contributes to the resistance of cells towards environmental stresses and killing effects of antibiotics, chemical biocides, disinfectants, serum immune components and immune cells. EPS acts as the cementing material of cells to adhere on the host surface and it helps to establish persistent biofilm colony in the host system [63]. In this study, EPS was extracted and dry weight was compared. The results delineated its significant reduction upto 52% in VA-treated cultures (Fig. 5b). Inhibition of surfactant production by VA.

Surfactant production is yet another virulence factor that is linked with QS-controlled swarming in many bacteria [64]. Microbial surfactant production is mostly controlled by QS and it promotes swarming motility by reducing the surface tension [65]. Surfactants help pathogens to consume lipids as nutrients and establish in a lipid-rich niche. Therefore, inhibition of surfactant production upon treatment with VA was measured. It has been noticed that upto 60% of surfactant production was inhibited (Fig. 4b). Thus, VA hampers the inhabitancy of *Y. enterocolitica* in a lipid-rich niche.

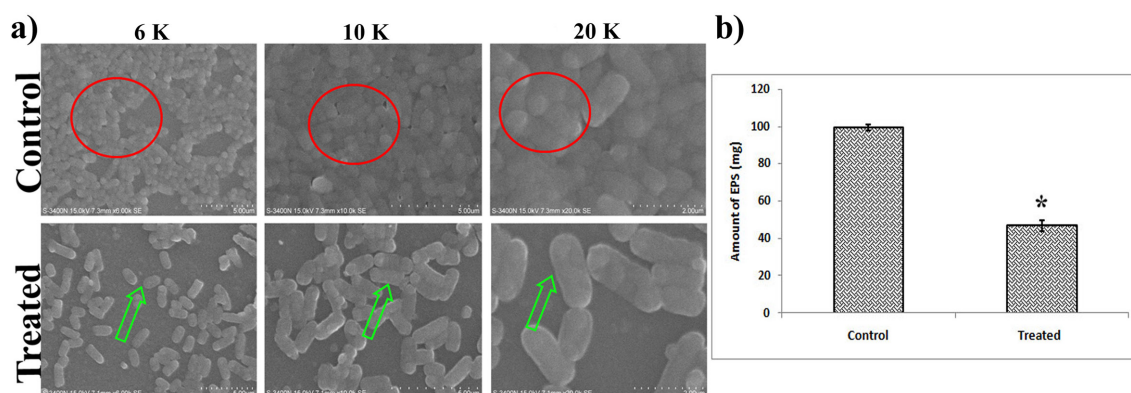


Fig. 5. (a) SEM image of control and treated cells at different magnifications such as 6000, 10000, 20000 $\times$. (b) Effect of vanillic acid on EPS production of *Y. enterocolitica*. Error bars and asterisks indicate the SD and statistical significance ($P \leq 0.05$), respectively.

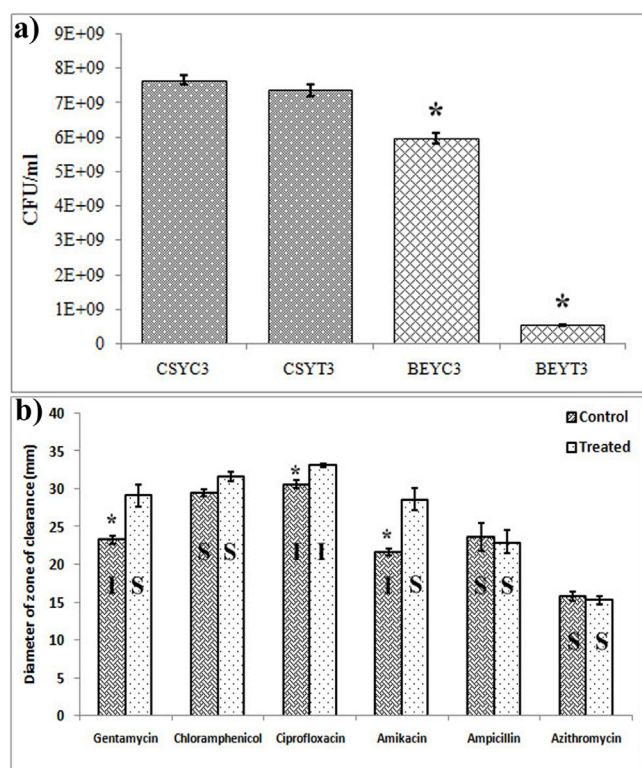


Fig. 6. (a) Effect of vanillic acid on susceptibility of *Y. enterocolitica* cells to human blood. Error bars and asterisks indicate the SD and statistical significance ($P \leq 0.05$), respectively. (b) Effect of tannic acid on susceptibility of *Y. enterocolitica* cells to antibiotics by disc diffusion method. Error bars and asterisks indicate the SD and statistical significance ($P \leq 0.05$), respectively.

Inhibition of blood resistivity by VA

As reported previously, *Y. enterocolitica* has the potential to cause bacteraemia and it has well enough viability in human blood. Fabrega *et al.* and China *et al.* [17, 66] have reported the role of YadA, an outer membrane protein encoded by virulence plasmid pYV against phagocytosis by human polymorphonuclear leukocytes. YopH (protein tyrosine phosphatase), YopE (cytotoxin) and YadA (adhesin) were identified to be involved in the endurance of *Y. enterocolitica* in host tissue against immune-cell attack [67]. Eradication of *Y. enterocolitica* from blood is vital to cure and alleviate the disease condition. Hence, the comparative sensitivity of untreated and VA treated *Y. enterocolitica* cells towards human blood were assessed by blood sensitivity assay. Upon blood exposure, the viability of control cells has been reduced nearly upto 22%, whereas, in VA-treated cells 92% of cells were killed. The results state that the resistance of *Y. enterocolitica* cells towards blood has got reduced by VA for about 91% (Fig. 6a).

Inhibition of antibiotic resistance by VA

Antibiotics are the prime and ultimate therapeutic regimen to treat *Y. enterocolitica*, however, the increasing drug resistance

makes it very complicated to treat patients. Abdel-Haq *et al.* [15] have documented the high degree of antibiotic resistance in *Y. enterocolitica* from children. Frazao *et al.* [68] have reported the intrinsic antibiotic resistance of *Y. enterocolitica*. A high proportion of isolates from pigs and pork were found to be resistant to antibiotics [18]. To test the effect of VA on antimicrobial sensitivity, the disc diffusion method was performed. The tested isolate was intermediate resistant to gentamycin, amikacin with a zone of clearance of 23 and 21 mm respectively, whereas by VA treatment it has increased upto 29 and 28 mm, respectively, which indicates the increased susceptibility of *Y. enterocolitica* towards the antibiotics. Thus, it has shifted the sensitivity type from intermediate resistant to sensitive. Similarly, it has slightly increased the sensitivity towards ciprofloxacin, where the zone of clearance was increased from 30 to 33 mm but sensitivity type did not change. On the other hand, no considerable variation was noticed in treatments with chloramphenicol, ampicillin and azithromycin (Fig. 6b). Further, the potential of VA to ameliorate the efficacy of antibiotics has been tested by broth micro-dilution assay to substantiate the disc diffusion assay. It has been shown that VA has potentiated the gentamycin, chloramphenicol, ciprofloxacin and amikacin, whereas in the case of ampicillin and azithromycin no change was observed (Fig. S2). Overall, VA has the potential to sensitize persistent cells towards antibiotics.

SEM analysis

The SEM analysis depicts that the control cells are totally encapsulated with EPS, whereas treated cells are defectively covered. The same was observed in all three magnifications such as 6000, 10000 and 20000 $\times$ (Fig. 5a). The SEM results correlated well with EPS quantification, in which significant inhibition of EPS production was observed (Fig. 5a). Thus, SEM analysis further substantiated the postulation that, the weaker encapsulation of treated cells by EPS, higher the sensitivity towards immune cells and antibiotics.

Differential gene-expression analysis

Total RNA was isolated from control and treated samples and their quality was assessed by 1% agarose gel electrophoresis (Fig. S3) and nanodrop (Table S1). As can be seen from the results (Fig. S3, Table S1), the RNA sample was qualified and taken for transcriptomic analysis. Overall, 4168 protein coding genes are present in *Y. enterocolitica* subsp. *enterocolitica* 8081 genome, out of those 3118 gene expressions were detected. The differential gene expression was depicted in scatter plot, volcano plot and heatmap (Fig S4a, b and S7). The important differentially expressed genes and their FPKM values were tabulated in Table 1. Among the most differentially expressed genes, many are correlating with the microbiological assay results (Fig. 7). A gene *sodB* is a stress response gene that codes for superoxide dismutase enzyme. It protects the cell from oxidative damage by endogenously produced free radicals; further SODs play a vital role in intraphagocytic survival of *Y. enterocolitica* [69]. VA treatment has augmented stress in *Y. enterocolitica* cells, as a consequence the stressed cells

Table 1. FPKM values of important differentially expressed genes

S.No	Gene	Function	FPKM_C	FPKM_T	S.No	Gene	Function	FPKM_C	FPKM_T
1	hslS	Heat shock chaperone IbpB	31.30	604.12	32	rscF	Outer membrane lipoprotein	54.30	33.99
2	sodB	Superoxide dismutase	36.95	518.58	33	emrA	Multidrug resistance protein A	13.82	6.11
3	pspA	Phage-shock-protein A	11.87	158.13	34	emrB	Multidrug resistance protein B	19.91	9.45
4	pspC	Phage shock protein	71.35	644.49	35	fliJ	Flagellar biosynthesis chaperone	33.66	0
5	pspB	Phage shock protein	139.35	778.68	36	fliL	Flagellar basal body-associated protein FliL	22.88	0
6	uspA	Universal stress protein A	279.15	2917.20	37	fliM	Flagellar motor switch protein FliM	37.43	5.55
7	acpP	Acyl carrier protein	329.69	3922.79	38	fliN	Flagellar motor switch protein FliN	37.43	5.55
8	hdeB	Persistence	72.73	783.42	39	fliO	Flagellar protein FliO	37.43	5.55
9	ompH	Outer membrane protein	64.33	656.02	40	fliZ	Flagella biosynthesis protein FliZ	22.22	4.88
10	cspE2	Cold shock protein CspE2	147.30	718.31	41	fliA	Flagellar biosynthesis sigma factor	15.87	5.90
11	cspC1	Cold shock-like protein CspC	597.82	4834.01	42	fliB	Flagellin lysine-N-methylase	14.89	4.26
12	cspB	Cold shock-like protein CspG	255.45	0	43	fliE	Flagellar hook-basal body protein FliE	43.01	0
13	cspD	Cold shock-like protein	264.84	2277.53	44	flgA	Flagellar basal body P-ring biosynthesis protein FlgA	6.31	0
14	cspA2	Major cold shock protein	9784.27	24666.30	45	flgN	Flagella synthesis protein FlgN	12.71	0
15	cspC2	Cold shock-like protein CspC2	833.21	3659.75	46	flgL	Flagellar hook-associated protein FlgL	16.34	0
16	sodA	Superoxide dismutase	44.44	238.45	47	flgB	Flagellar basal body rod protein FlgB	24.95	0
17	napA	Periplasmic nitrate reductase	82.16	9.43	48	flgJ	Flagellar rod assembly protein/muramidase FlgJ	16.43	1.30
18	napD	Chaperone NapD	82.16	9.43	49	flgF	Flagellar basal body rod protein FlgF	15.40	3.85
19	napF	Ferredoxin-type protein NapF	82.16	9.43	50	flgE	Flagellar hook protein FlgE	13.11	4.01
20	foxA	Ferrioxamine receptor	14.76	2.73	51	flgD	Flagellar basal body rod modification protein	21.38	10.85
21	fepC	Iron-siderophore ABC transporter ATP-binding protein	51.43	12.78	52	flhA	Flagellar biosynthesis protein FlhA	37.82	5.07
22	fepD	Iron-enterobactin transporter membrane protein	51.43	12.78	53	flhB	Flagellar biosynthesis protein FlhB	37.82	5.07
23	fepG	Iron-enterobactin transporter permease	51.43	12.78	54	flhE	Flagellar protein FlhE	37.82	5.07
24	feoA	Ferrous iron transport protein A	47.55	0	55	flhC	Transcriptional activator FlhC	19.95	7.73

Continued

Table 1. Continued

S.No	Gene	Function	FPKM_C	FPKM_T	S.No	Gene	Function	FPKM_C	FPKM_T
25	myfE	Regulatory protein	24.30	18.61	56	flhD	Transcriptional activator FlhD	48.35	14.41
26	myfF	Hypothetical protein	24.30	18.61	57	motA	Flagellar motor protein MotA	28.77	5.96
27	myfC	Outer membrane usher protein	3.13	0.45	58	motB	Flagellar motor protein MotB	28.77	5.96
28	yegB	Multidrug efflux system protein MdtE	16.21	2.80	59	cheB	Chemotaxis-specific methylesterase	30.90	14.14
29	yegM	Multidrug efflux system subunit MdtA	38.48	15.28	60	cheR	Chemotaxis methyltransferase CheR	30.90	14.14
30	yegN	Multidrug efflux system subunit MdtB	38.48	15.28	61	ugpC	sn-glycerol-3-phosphate import ATP-binding protein UgpC	41.23	4.07
31	yegO	Multidrug efflux system subunit MdtC	38.48	15.28	62	ugpQ	Glycerophosphodiester phosphodiesterase	41.23	4.07

Vanillic acid tames QS in *Y. enterocolitica*

CSH, Cell surface hydrophobicity; EPS, Extra Polymeric Substances; MATH, Microbial Adherence to Hydrocarbons; MQIC, Minimal Quorum Inhibitory Concentration; QS, Quorum sensing; SEM, Scanning Electron Microscope; VA, Vanillic acid.

provoked the expression of stress-response genes such as *sodB* and *sodA* to ensure its survival. Parallely, *hslS* gene codes for small heat-shock protein *IbpB*, which was also up regulated. It is a stress-response chaperone that aids in the refolding of denatured proteins along with other chaperones. It stabilizes aggregation-prone folding intermediates of denatured proteins during stress, an integral part of a cooperative multi-chaperone network [70]. The *pspA* is another stress-response gene that has got up regulated significantly. Along with that *pspC* and *pspB* were also induced considerably. The phage-shock-protein system (*Psp*) helps the bacteria to endure during dissipation of the proton motive force (PMF) and is conserved in many Gram-negative bacteria. The expression of the *Psp* system has been linked to important bacterial phenotypes including virulence in THE host system and survival within macrophages [71]. On the whole, the *psp* operon helps the bacteria to tackle the stress and survive better. In accordance with that, VA treatment has provoked the expression of *psp* genes, which indicates that the cells are trying to get rid of stress introduced by VA treatment. The *uspA* is yet another gene involved in recovery of cells from stress. It expresses upon a broad range of stresses such as growth retardation, free radical exposure, osmotic stress etc., so that it is designated as universal stress response gene [72]. Upon VA treatment, the *uspA* level has also been noticeably increased, which indicates the imposition of stress by VA.

Acyl carrier protein gene (*acpP*) was found to be elicited upon VA treatment. The role of *acpP* of *Pseudomonas aeruginosa* was reported as membrane fatty acid synthesis gene. It is also closely related to *acp* gene series, which are stress-response genes [73]. The *hdeB* is one of the significantly up-regulated genes upon VA treatment. It is an acid-resistance stress-response gene in enteric pathogens, which facilitates the

survival in high acidic environment [74]. As a consequence of VA treatment these stress-response genes are up regulated to rescue the cells by increasing the resistivity against stress. The *ompH* gene expresses an outer-membrane protein involved in dispersal of hydrophilic solutes through the outer membrane. It is highly immunogenic in nature and it has been established as a vaccine target to treat many bacterial infections through immunization. VA has significantly induced the expression of *ompH*, as a consequence it has increased the immunogenicity of the cells by provoking the opsonization of *Y. enterocolitica* cells by immune cells. In the present study, through blood sensitivity assay it has been found that VA has inhibited the survival of *Y. enterocolitica* cells in live human blood. The cold-shock proteins such as *cspE2*, *cspC1*, *cspD*, *cspA2*, *cspC2* are considerably over expressed. The *csp* gene series are facilitated to tolerate cold-temperature stress, moreover, it is also involved in other stresses such as oxidative, osmotic, starvation, pH and ethanol stress tolerance [75]. This further indicates that, VA provokes broad-spectrum of stress in *Y. enterocolitica*.

The *nap* operon is mainly comprised of *napA*, *napD* and *napF*, which are involved in nitrate reduction. Nitrate is the electron acceptor in anaerobic condition; *nap* operon facilitates survival and growth in anaerobic condition [76]. Upon VA treatment, *napADF* are significantly down regulated, thus the endurance of *Y. enterocolitica* in anerobic condition will be diminished. The *foxA* gene was significantly suppressed in treated *Y. enterocolitica*, which determines the intake of ferric ion. Synthesis of siderophore transport system for iron is one of the key virulence factors in *Y. enterocolitica* infection. The *foxA* gene was found to be highly conserved among the pathogenic strains [77]. Similarly, *fepD*, *fepG* and *fepC* have also been diminished upon VA treatment. The *fepDGC*

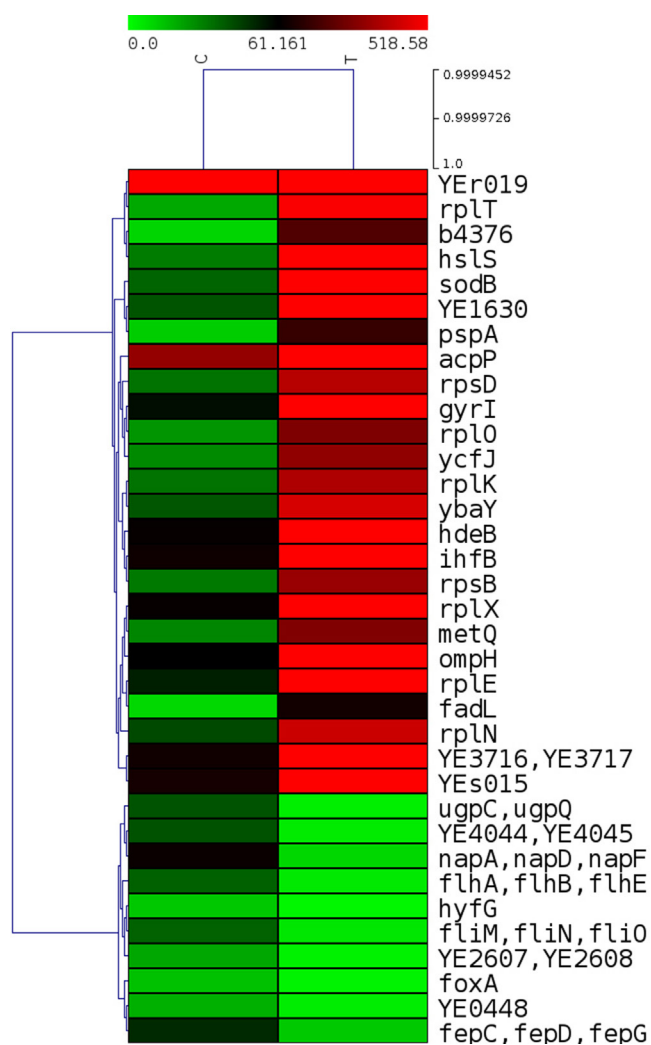


Fig. 7. Heatmap of topmost differentially expressed genes between control and treated *Y. enterocolitica* cells.

codes for iron-enterobactin transporter membrane protein, transporter permease and iron-siderophore ABC transporter ATP-binding protein, respectively. These genes are involved in ferrous ion (Fe^{2+}) metabolism, which is essential for facultative anaerobic micro-organisms to sustain iron-restricted environment [78]. Therefore, VA treatment reduces the sustainability in iron-restricted environment and also had reduced the fep-associated virulence. Moreover, feoA has been found to be drastically suppressed in VA treatment, while it was expressed only in control. It is acting as mandatory supporting protein for feoB for iron uptake. It has been recognized as a virulence factor in many Gram-negative bacteria [79]. The genes myfE, myfF and myfG codes for surface antigens and they closely resemble enterotoxin-associated fimbriae of many other Gram-negative bacteria. Myf works as a virulence factor in conjunction with Yst to cause diarrhoea. These genes were found to be considerably reduced in VA-treated cells [80].

Huge numbers and diverse varieties of multidrug-efflux systems exist in bacteria to protect cells from different environmental toxins and noxious compounds by active extrusion. The transporter gene cluster called yegMNOB also known as mdtABCD is one such major multidrug-efflux transporters in Gram-negative bacteria [81]. This gene cluster has been down regulated by VA treatment. This repression of drug-efflux system resulted in a significant reduction of resistance against antibiotics. The genes emrA and emrB are multidrug-resistance proteins [82], which are also found to be diminished by VA treatment. As can be seen from the result (Fig. 6b and S2) VA-treated cells are more susceptible to gentamicin, chloramphenicol, ciprofloxacin and amikacin. VA has also inhibited the expression of rcsF gene, which is a part of Rcs phosphorelay system. The rcs system plays a vital role in systemic infection, importantly against phagocytic immune cells [83]. It also positively regulates the ysa-tsp T3SS genes involved in early stage infection in the host. The suppression of rcsF would also be a cause for increased sensitivity of *Y. enterocolitica* cells towards immune cells (Fig. 6a).

The fliJ, fliL and fliE genes correspond to the flagellar biosynthesis chaperone. In our study, it was found that the flagellar structural proteins expressed by fliJ, fliL, fliE, flgA, flgN, flgL and flgB genes [84] are not expressed in VA-treated cells and expressed only in controls. Motility of *Y. enterocolitica* is highly associated and directly proportional to virulence factor and it can be even taken as a therapeutic target [85]. Moreover, fliM, fliN, fliO, fliZ, fliA and fliB that code for flagellar motor switch protein, flagellar structural protein, flagellar biosynthesis sigma factor and flagellin lysine-N-methylase, respectively, are considerably down regulated. Some other motility genes such as flgA, flgN, flgL and flgB [84] were not detected in treated and found only in control cells. Flagellar biosynthesis proteins viz., flhA, flhB and flhE were significantly down regulated. The flgJ, flgF, flgE, flgD, flhC, flhD, motA and motB encoding flagellar rod assembly proteins, structural proteins, transcriptional activator, flagellar motor proteins motA and B [84], respectively, were also found considerably reduced. The cheB and cheR genes were suppressed along with flagella-related genes, which encode for chemotaxis-specific methyltransferase and chemotaxis methyltransferase, respectively, which are involved in chemotaxis. It helps bacteria to recognize the chemical composition of its environment and move accordingly. These genes are highly associated with motility. These flagella and chemotaxis-related genes are directly proportional to biofilm development by *Y. enterocolitica*. Hence, declension of these gene expressions may retard the biofilm formation in *in vivo* condition and thus VA can reduce the pathogenicity and disease progression.

The genes ugpC and ugpQ that encode sn-glycerol-3-phosphate (G3P) import ATP-binding protein and glycerophosphodiester phosphodiesterase, respectively, are significantly inhibited by VA treatment. These genes are involved in utilization of G3P, a degradation product of phospholipids as carbon and phosphate source under starvation condition. Indirectly, ugpC and ugpQ helps bacterial cells to ingest a variety of nutrients, osmoprotectants, diverse growth

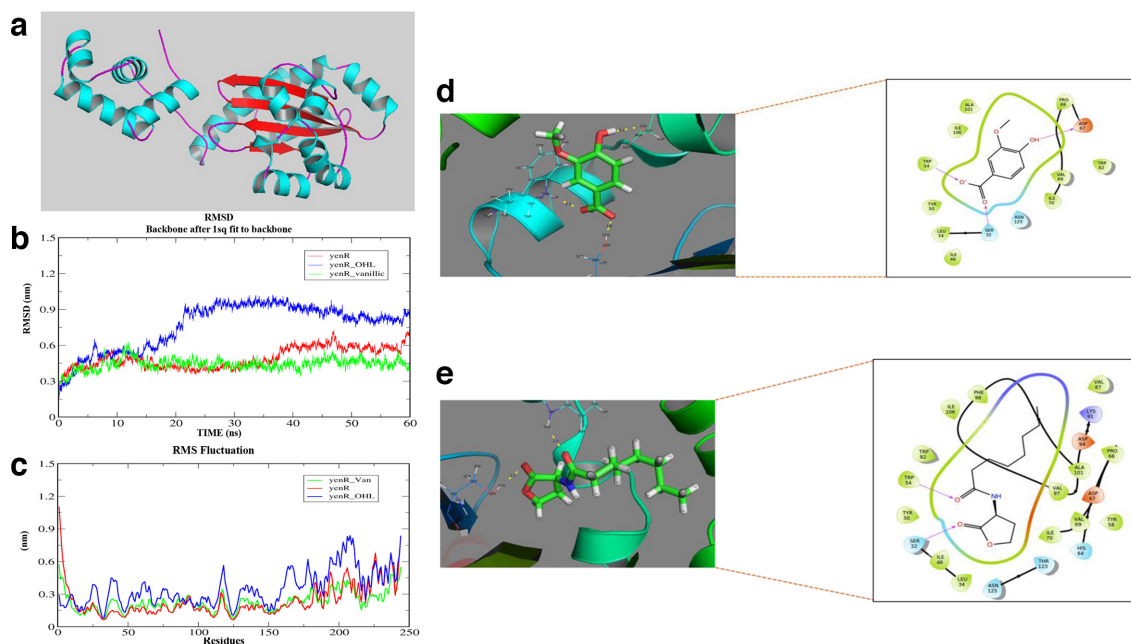


Fig. 8. (a) Representative view of the modelled yenR protein; (cyan, red and pink colour represents alpha helices, beta sheets and coils, respectively), (b) RMSD plot for backbone of yenR (red), yenR-vanillic acid (green) and yenR-OHL complex (blue), (c) RMSF plot for C-alpha residues for yenR and its complex with vanillic acid and blue [colouring scheme is the same as in (b)]. (d) Docked conformation of yenR with vanillic acid, (e) docked conformation of yenR with OHL; yellow dashed lines (left-3D presentation) and pink arrow (right-2D representation) showing hydrogen bond interaction.

factors and trace elements. They also play a vital role in maintaining cell integrity, environmental stress response, cell-to-cell communication, cell differentiation and pathogenicity [86]. Thus, VA seems to introduce yet another stress to *Y. enterocolitica* cells by suppressing these genes and reduce their sustainability in host environment. All the identified genes corresponding to proteins were categorized and tabulated as Table S3 in XL file.

Molecular simulation and docking analysis

In silico analysis, such as molecular simulation, docking and molecular dynamics were done to further corroborate the anti-QS activity of VA. YenR is a transcriptional activator of QS system of *Y. enterocolitica*, which consists of 244 residues. The crystal structure of yenR (PDB ID- 5LO7) was reported by [40]. However, it had an electron density of 179 (of 244) residues. Hence, yenR was modelled by multi-template homology using modeller 9.22 (Fig. 8a). Ramachandran plot analysis (PROCHECK) of the modelled yenR structure showed that 90% of the structure is in the favoured region (Table S2) and ProSA-web server calculation showed a better Z score of -7.8 than the template (Fig. S5a). SAVES server analysis and ERRAT analysis also gave a very good score of 82–84% for modelled yenR.

The molecular docking study reveals the interaction of yenR with two ligands (Fig. S6a, b). Several active site pockets were derived from CASTp out of which one pocket was chosen for docking study using Schrödinger-Maestro 2018.1. The

ligand preparation was done by LIGPREP followed by energy minimization and pre-processing of protein by protein wizard [44]. Table 2 describes the binding energy, hydrogen-bonding interactions of docked complexes derived after docking.

Molecular dynamics analysis was done to test the stability of yenR-VA and yenR-OHL complexes. Root-mean-square deviation (RMSD) plot of the modelled structure exhibited some stabilization between 70 to 80 ns. Also, root-mean-square fluctuation (RMSF) plot indicated a considerable fluctuation of residues at C-terminal (180-244) (Figs 8 and

Table 2. Interaction of yenR with vanillic acid and N-octanoyl homoserine lactone (OHL)

Interaction of yenR	Hydrogen-bonding interactions Residue (bond length Å)	Binding energy (kcal/mol)	Glide-Emodel score	Residues showing hydrophobic interactions
Vanillic Acid (8468)	32SER (1.8), 54TRP (2.1), 67ASP (3.21)	-6.30	-36.731	Pro68, Val69, Ile70
OHL (6914579).	32SER(2.1), 54 TRP(2.0)	-5.40	-30.537	Pro68, Val69, Ile70, 82Trp, 97Val 101Ala, 106Ile

S5). However, it was observed that the YenR has attained stability upon binding with VA (green coloured in Fig. 8b, c). The yenR-VA exhibited stable deviation after 40 ns. It was found that during the last 20 ns of simulation period, yenR-VA showed a stable deviation with RMSD of about 0.3 and 0.5 nm whereas, yenR-OHL complex and yenR protein was about 1 and 0.65 nm, respectively. The RMSF profiles were also calculated for the protein complexes during simulation (Fig. 8c). The yenR residues in yenR-VA complex showed a high degree of stability throughout simulation of 60 ns, whereas, yenR in the yenR-OHL complex showed much more fluctuations. Table 2 showed the binding energy of yenR with corresponding ligands. As mentioned in the table, the yenR+vanillic acid complex had considerably more binding energy (−6.30 kcal/mol) as compared to yenR+OHL complex (−5.40 kcal/mol). Also, vanillic acid had shown three hydrogen bonds and three hydrophobic interactions with yenR, compared to two hydrogen bonds as observed in yenR+OHL complex. This difference in the number of hydrogen bonds can also be playing their role in stabilizing the complex as seen in RMSD and RMSF profiles (Fig. 8b, c). The superimposed structures of the respective protein or protein+ligand complex in the intervals of 10 ns were showed in Figs S7–S9. The representative views of the protein in 10 ns in juxtaposition are also shown. A short video is also attached as Supplemental Videos (SV1, SV2 and SV3). Thus, the result states that VA bound better than that of default signalling molecule OHL to the QS receptor protein. The stronger interaction between VA and YenR may lead to the competitive inhibition of OHL-YenR, thus quenches QS signals. Therefore, it can be an ideal inhibitor for yenR protein to quench QS.

In summary, the present study deciphers the anti-QS activity and anti-pathogenic activity of VA against *Y. enterocolitica*. It has inhibited the motility, CSH, EPS production and surfactant production. Moreover, it has significantly sensitized the cells towards antibiotics and human immune cells. Thus, VA has improved the treatability of *Y. enterocolitica* associated infections. In furtherance, transcriptomic analysis has revealed the down regulation of virulence-related genes and up regulation of various stress-response genes. The anti-QS activity was confirmed by *in silico* analysis such as molecular docking and molecular dynamics. Thus the quorum quenching and anti-pathogenic potential of VA against *Y. enterocolitica* has been clearly demonstrated.

Funding information

This study was supported by Science and Engineering Research Board, Government of India, through National Post Doctoral Fellowship (N-PDF) to Dr C. Sivasankar [Sanction number PDF/2016/004041]. The study was also partly supported by Research Associateship to Dr C. Sivasankar [File No: 3/1/2/1/Env/2019-NCD-1] funded by Indian Council of Medical Research, India.

Author contributions

C.S.S. designed the study, performed the experiments, analysed the data, prepared the figures and tables and wrote the manuscript. N.K.J. performed the experiments, analysed the data. S.R.S. performed the experiments, analysed the data prepared the figures and tables for the *in silico* part of the study. A.M. designed and supervised the *in silico*

part of the study. P.K.S. designed and supervised the study. C.S.S. and N.K.J. equally contributed as first authors. All authors have read and approved the final version of the manuscript.

Conflicts of interest

The authors declare that there are no conflicts of interest.

Ethical statement

This article does not contain any studies with animals performed by any of the authors. The blood sensitivity assay was performed by taking blood from healthy human participant. Other than extracting blood, human subjects were not involved in any other way of the present study. The experiment was done with an appropriate ethical clearance from the Human Ethical Committee, Pondicherry University (No: PU/FST/HEC/2018-19/17). All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all individual participants included in the study.

References

- Ng Y-K, Grasso M, Wright V, Garcia V, Williams P et al. The quorum sensing system of *Yersinia enterocolitica* 8081 regulates swimming motility, host cell attachment, and virulence plasmid maintenance. *Genes* 2018;9:307.
- Vanantwerpen G, Houf K, Van Damme I, Berkvens D, De Zutter L. Estimation of the within-batch prevalence and quantification of human pathogenic *Yersinia enterocolitica* in pigs at slaughter. *Food Control* 2013;34:9–12.
- Gopu V, Shetty PH. Regulation of acylated homoserine lactones (AHLs) in beef by spice marination. *J Food Sci Technol* 2016;53:2686–2694.
- Sirghani K, Zeinali T, Jamshidi A. Detection of *Yersinia enterocolitica* in Retail Chicken Meat, Mashhad, Iran. *J Pathog* 2018;2018:1–4.
- Li C, Gözl G, Alter T, Barac A, Hertwig S et al. Prevalence and antimicrobial resistance of *Yersinia enterocolitica* in retail seafood. *J Food Prot* 2018;81:497–501.
- Verbikova V, Borilova G, Babak V, Moravkova M. Prevalence, characterization and antimicrobial susceptibility of *Yersinia enterocolitica* and other *Yersinia* species found in fruits and vegetables from the European Union. *Food Control* 2018;85:161–167.
- Petsios S, Fredriksson-Ahomaa M, Sakkas H, Papadopoulos C. Conventional and molecular methods used in the detection and subtyping of *Yersinia enterocolitica* in food. *Int J Food Microbiol* 2016;237:55–72.
- Koehler KM, Lasky T, Fein SB, Delong SM, Hawkins MA et al. Population-based incidence of infection with selected bacterial enteric pathogens in children younger than five years of age, 1996–1998. *Pediatr Infect Dis J* 2006;25:129–134.
- Lamps LW, Madhusudhan KT, Greenson JK, Pierce RH, Massoli NA et al. The role of *Yersinia enterocolitica* and *Yersinia pseudotuberculosis* in granulomatous appendicitis: a histologic and molecular study. *Am J Surg Pathol* 2001;25:508–515.
- Abraham M, Pai M, Kang G, Asokan GV, Magesh SR et al. An outbreak of food poisoning in Tamil Nadu associated with *Yersinia*. *Indian J Med Res* 1997;106:465–468.
- Bucher M, Meyer C, Grötzbach B, Wacheck S, Stolle A et al. Epidemiological data on pathogenic *Yersinia enterocolitica* in Southern Germany during 2000–2006. *Foodborne Pathog Dis* 2008;5:273–280.
- Butler T, Islam M, Islam MR, Azad AK, Huq MI et al. Isolation of *Yersinia enterocolitica* and *Y. intermedia* from fatal cases of diarrhoeal illness in Bangladesh. *Trans R Soc Trop Med Hyg* 1984;78:449–450.
- Okwori AEJ, Martínez PO, Fredriksson-Ahomaa M, Agina SE, Korkeala H. Pathogenic *Yersinia enterocolitica* 2/0:9 and *Yersinia pseudotuberculosis* 1/0:1 strains isolated from human and non-human sources in the plateau state of Nigeria. *Food Microbiol* 2009;26:872–875.

14. Gupta V, Gulati P, Bhagat N, Dhar MS, Viridi JS. Detection of *Yersinia enterocolitica* in food: an overview. *Eur J Clin Microbiol Infect Dis* 2015;34:641–650.
15. Abdel-Haq NM, Papadopol R, Asmar BI, Brown WJ. Antibiotic susceptibilities of *Yersinia enterocolitica* recovered from children over a 12-year period. *Int J Antimicrob Agents* 2006;27:449–452.
16. Peng Z, Zou M, Li M, Liu D, Guan W et al. Prevalence, antimicrobial resistance and phylogenetic characterization of *Yersinia enterocolitica* in retail poultry meat and swine feces in parts of China. *Food Control* 2018;93:121–128.
17. Fàbrega A, Vila J. *Yersinia enterocolitica*: pathogenesis, virulence and antimicrobial resistance. *Enferm Infecc Microbiol Clin* 2012;30:24–32.
18. Martins BTF, Botelho CV, Silva DAL, Lanna F, Grossi JL et al. *Yersinia enterocolitica* in a Brazilian pork production chain: Tracking of contamination routes, virulence and antimicrobial resistance. *Int J Food Microbiol* 2018;276:5–9.
19. Atkinson S, Chang CY, Sockett RE, Cámara M, Williams P. Quorum sensing in *Yersinia enterocolitica* controls swimming and swarming motility. *J Bacteriol* 2006;188:1451–1461.
20. Truchado P, Tomás-Barberán FA, Larrosa M, Allende A. Food phytochemicals act as quorum sensing inhibitors reducing production and/or degrading autoinducers of *Yersinia enterocolitica* and *Erwinia carotovora*. *Food Control* 2012;24:78–85.
21. Giménez-Bastida JA, Truchado P, Larrosa M, Espín JC, Tomás-Barberán FA et al. Urolithins, ellagitannin metabolites produced by colon microbiota, inhibit quorum sensing in *Yersinia enterocolitica*: phenotypic response and associated molecular changes. *Food Chem* 2012;132:1465–1474.
22. Lekshmi M, Parvathi A, Kumar S, Varela MF. Efflux pump-mediated quorum sensing: new avenues for modulation of antimicrobial resistance and bacterial virulence. *Biotechnological Applications of Quorum Sensing Inhibitors*. Singapore: Kalia VC. Springer; 2018. pp. 27–142.
23. Khatkar A, Sharma KK. Phenylpropanoids and its derivatives: biological activities and its role in food, pharmaceutical and cosmetic industries. *Crit Rev Food Sci Nutr* 2020;60:1–21.
24. Santhakumari S, Jayakumar R, Logalakshmi R, Prabhu NM, Abdul Nazar AK et al. *In vitro* and *in vivo* effect of 2,6-Di-tert-butyl-4-methylphenol as an antibiofilm agent against quorum sensing mediated biofilm formation of *Vibrio* spp. *Int J Food Microbiol* 2018;281:60–71.
25. Li Y, Bai F, Xia H, Zhuang L, Xu H et al. A novel regulator PA5022 (aefA) is involved in swimming motility, biofilm formation and elastase activity of *Pseudomonas aeruginosa*. *Microbiol Res* 2015;176:14–20.
26. CLSI. *Performance Standards for Antimicrobial Disk Susceptibility Tests; Approved Standard—Eleventh Edition*. CLSI Document M02-A11. Wayne, PA: Clinical and Laboratory Standards Institute; 2012.
27. Selvaraj A, Jayasree T, Valliammai A, Pandian SK. Myrtenol attenuates MRSA biofilm and virulence by suppressing *sarA* expression dynamism. *Front Microbiol* 2019;10:10.
28. Sivasankar C, Maruthupandian S, Balamurugan K, James PB, Krishnan V et al. A combination of ellagic acid and tetracycline inhibits biofilm formation and the associated virulence of *Propionibacterium acnes* in vitro and in vivo. *Biofouling* 2016;32:397–410.
29. Viszwapiya D, Prithika U, Deebika S, Balamurugan K, Pandian SK. *In vitro* and *in vivo* antibiofilm potential of 2,4-Di-tert-butylphenol from seaweed surface associated bacterium *Bacillus subtilis* against group A Streptococcus. *Microbiol Res* 2016;191:19–31.
30. Bakkiyaraj D, Sivasankar C, Pandian SK. Inhibition of quorum sensing regulated biofilm formation in *Serratia marcescens* causing nosocomial infections. *Bioorg Med Chem Lett* 2012;22:3089–3094.
31. Padmavathi AR, Pandian SK. Antibiofilm activity of biosurfactant producing coral associated bacteria isolated from Gulf of Mannar. *Indian J Microbiol* 2014;54:376–382.
32. Sivasankar C, Ponmalar A, Bhaskar JP, Pandian SK. Glutathione as a promising anti-hydrophobicity agent against *Malassezia* spp. *Mycoses* 2015;58:620–631.
33. Sivarajani M, Leskinen K, Aravindraja C, Saavalainen P, Pandian SK et al. Deciphering the antibacterial mode of action of alpha-Mangostin on *Staphylococcus epidermidis* RP62A through an integrated transcriptomic and proteomic approach. *Front Microbiol* 2019;10:150.
34. Bi L, Guan CJ, Yang GE, Yang F, Yan HY et al. High-throughput transcriptome sequencing analysis provides preliminary insights into the biotransformation mechanism of *Rhodospseudomonas palustris* treated with alpha-rhamnetin-3-rhamnoside. *Microbiol Res* 2016;185:1–12.
35. Laskowski RA, MacArthur MW, Moss DS, Thornton JM. PROCHECK: a program to check the stereochemical quality of protein structures. *J Appl Crystallogr* 1993;26:283–291.
36. Oostenbrink C, Villa A, Mark AE, van Gunsteren WF. A biomolecular force field based on the free enthalpy of hydration and solvation: the GROMOS force-field parameter sets 53A5 and 53A6. *J Comput Chem* 2004;25:1656–1676.
37. van Aalten DMF, van Aalten DMF. PRODRG: a tool for high-throughput crystallography of protein-ligand complexes. *Acta Crystallogr D Biol Crystallogr* 2004;60:1355–1363.
38. Friesner RA, Murphy RB, Repasky MP, Frye LL, Greenwood JR et al. Extra precision glide: docking and scoring incorporating a model of hydrophobic enclosure for protein-ligand complexes. *J Med Chem* 2006;49:6177–6196.
39. Wiederstein M, Sippl MJ. ProSA-web: interactive web service for the recognition of errors in three-dimensional structures of proteins. *Nucleic Acids Res* 2007;35:W407–W410.
40. Kim Y, Chhor G, Jedrzejczak R, Winans SC, Anderson WF et al. A Crystal structure of quorum-sensing transcriptional activator from *Yersinia enterocolitica*. Center for Structural Genomics of Infectious Diseases (CSGID) 2016.
41. Webb B, Sali A. Comparative protein structure modeling using MODELLER. *Curr Protoc Bioinformatics* 2016;54:5.6.1–5.6.5.
42. Tian W, Chen C, Lei X, Zhao J, Liang J. CASTp 3.0: computed atlas of surface topography of proteins. *Nucleic Acids Res* 2018;46:W363–W367.
43. Maestro. 1. Maestro. New York, NY: Schrödinger, LLC; 2018.
44. Schrödinger L. *Schrödinger Software Suite*. New York: Schrödinger, LLC; 2018.
45. Abraham MJ, van der Spoel D, Lindahl E, Hess B. The GROMACS development team. GROMACS user manual version 2019.
46. Rajkumari J, Borkotoky S, Reddy D, Mohanty SK, Kumavath R et al. Anti-quorum sensing and anti-biofilm activity of 5-hydroxymethylfurfural against *Pseudomonas aeruginosa* PA01: insights from in vitro, in vivo and in silico studies. *Microbiol Res* 2019;226:19–26.
47. Singh D, Rawat S, Waseem M, Gupta S, Lynn A et al. Molecular modeling and simulation studies of recombinant laccase from *Yersinia enterocolitica* suggests significant role in the biotransformation of non-steroidal anti-inflammatory drugs. *Biochem Biophys Res Commun* 2016;469:306–312.
48. Ahlawat S, Singh D, Yadav A, Singh AK, Viridi JS et al. Proteomic analysis reveals the damaging role of low redox laccase from *Yersinia enterocolitica* strain 8081 in the midgut of *Helicoverpa armigera*.
49. Kanan TA, Abdulla ZA. Isolation of *Yersinia* spp. from cases of diarrhoea in Iraqi infants and children. *East Mediterr Health J* 2009;15:276–284.
50. Rahman A, Bonny TS, Stonsaovapak S, Ananchaipattana C. *Yersinia enterocolitica*: epidemiological studies and outbreaks. *J Pathog* 2011;2011:1–11.
51. Fredriksson-Ahomaa M, Stolle A, Siitonen A, Korkeala H. Sporadic human *Yersinia enterocolitica* infections caused by bioserotype 4/O:3 originate mainly from pigs. *J Med Microbiol* 2006;55:747–749.
52. Chen X, Zhang L, Zhang M, Liu H, Lu P et al. Quorum sensing inhibitors: a patent review (2014–2018). *Expert Opin Ther Pat* 2018;28:849–865.

53. Gopu V, Meena CK, Shetty PH. Quercetin influences quorum sensing in food borne bacteria: *in-vitro* and *in-silico* evidence. *PLoS One* 2015;10:e0134684.
54. Truchado P, Giménez-Bastida J-A, Larrosa M, Castro-Ibáñez I, Espín JC et al. Inhibition of quorum sensing (Qs) in *Yersinia enterocolitica* by an orange extract rich in glycosylated flavanones. *J Agric Food Chem* 2012;60:8885–8894.
55. Bernal-Mercado AT, Vazquez-Armenta FJ, Tapia-Rodriguez MR, Islas-Osuna MA, Mata-Haro V et al. Comparison of single and combined use of catechin, protocatechuic, and vanillic acids as antioxidant and antibacterial agents against uropathogenic *Escherichia coli* at planktonic and biofilm levels. *Molecules* 2018;23:2813.
56. Gong J, Zhou S, Yang S. Vanillic acid suppresses HIF-1 α expression via inhibition of mTOR/p70S6K/4E-BP1 and Raf/MEK/ERK pathways in human colon cancer HCT116 cells. *Int J Mol Sci* 2019;20:465.
57. Bai F, Fang L, Hu H, Yang Y, Feng X, et al FX et al. Vanillic acid mitigates the ovalbumin (OVA)-induced asthma in rat model through prevention of airway inflammation. *Biosci Biotechnol Biochem* 2019;83:531–537.
58. Siddiqui S, Kamal A, Khan F, Jamali KS, Saify ZS. Gallic and vanillic acid suppress inflammation and promote myelination in an in vitro mouse model of neurodegeneration. *Mol Biol Rep* 2019;46:997–1011.
59. Doyle RJ. Contribution of the hydrophobic effect to microbial infection. *Microbes Infect* 2000;2:391–400.
60. Tamm A, Tarkkanen AM, Korhonen TK, Kuusela P, Toivanen P et al. Hydrophobic domains affect the collagen-binding specificity and surface polymerization as well as the virulence potential of the YadA protein of *Yersinia enterocolitica*. *Mol Microbiol* 1993;10:995–1011.
61. Lachica RV, Zink DL. Plasmid-associated cell surface charge and hydrophobicity of *Yersinia enterocolitica*. *Infect Immun* 1984;44:540–543.
62. Paerregaard A, Espersen F, Jensen OM, Skurnik M. Interactions between *Yersinia enterocolitica* and rabbit ileal mucus: growth, adhesion, penetration, and subsequent changes in surface hydrophobicity and ability to adhere to ileal brush border membrane vesicles. *Infect Immun* 1991;59:253–260.
63. Gunn JS, Bakaletz LO, Wozniak DJ. What's on the outside matters: the role of the extracellular polymeric substance of gram-negative biofilms in evading host immunity and as a target for therapeutic intervention. *J Biol Chem* 2016;291:12538–12546.
64. Köhler T, Curty LK, Barja F, van Delden C, Pechère JC. Swarming of *Pseudomonas aeruginosa* is dependent on cell-to-cell signaling and requires flagella and pili. *J Bacteriol* 2000;182:5990–5996.
65. Kearns DB. A field guide to bacterial swarming motility. *Nat Rev Microbiol* 2010;8:634–644.
66. China B, N'Guyen BT, De Bruyère M, Cornelis GR. Role of YadA in resistance of *Yersinia enterocolitica* to phagocytosis by human polymorphonuclear leukocytes. *Infect Immun* 1994;62:1275–1281.
67. Grosdent N, Maridonneau-Parini I, Sory MP, Cornelis GR. Role of Yops and adhesins in resistance of *Yersinia enterocolitica* to phagocytosis. *Infect Immun* 2002;70:4165–4176.
68. Frazão MR, Andrade LN, Darini AL, Falcão JP. Antimicrobial resistance and plasmid replicons in *Yersinia enterocolitica* strains isolated in Brazil in 30 years. *Braz J Infect Dis* 2017;21:477–480.
69. Dhar MS, Gupta V, Virdi JS. Detection, distribution and characterization of novel superoxide dismutases from *Yersinia enterocolitica* biovar 1A. *PLoS One* 2013;8:e63919.
70. Veinger L, Diamant S, Buchner J, Goloubinoff P. The small heat-shock protein IbpB from *Escherichia coli* stabilizes stress-denatured proteins for subsequent refolding by a multichaperone network. *J Biol Chem* 1998;273:11032–11037.
71. Gueguen E, Savitzky DC, Darwin AJ. Analysis of the *Yersinia enterocolitica* PspBC proteins defines functional domains, essential amino acids and new roles within the phage-shock-protein response. *Mol Microbiol* 2009;74:619–633.
72. Nyström T, Neidhardt FC. Expression and role of the universal stress protein, UspA, of *Escherichia coli* during growth arrest. *Mol Microbiol* 1994;11:537–544.
73. Chen W, Wang B, Gruber JD, Zhang YM, Davies C. Acyl Carrier Protein 3 is involved in oxidative stress response in *Pseudomonas aeruginosa*. *Front Microbiol* 2018;9:2244.
74. Hong W, Wu YE, Fu X, Chang Z. Chaperone-dependent mechanisms for acid resistance in enteric bacteria. *Trends Microbiol* 2012;20:328–335.
75. Li C, Murugaiyan J, Thomas C, Alter T, Riedel C. Isolate specific cold response of *Yersinia enterocolitica* in transcriptional, proteomic, and membrane physiological changes. *Front Microbiol* 2019;10:3037.
76. Nilavongse A, Brondijk THC, Overton TW, Richardson DJ, Leach ER et al. The NapF protein of the *Escherichia coli* periplasmic nitrate reductase system: demonstration of a cytoplasmic location and interaction with the catalytic subunit, NapA. *Microbiology* 2006;152:3227–3237.
77. Huang Y, Wang X, Cui Z, Yang Y, Xiao Y et al. Possible use of ail and foxA polymorphisms for detecting pathogenic *Yersinia enterocolitica*. *BMC Microbiol* 2010;10:211.
78. Schubert S, Fischer D, Heesemann J. Ferric enterochelin transport in *Yersinia enterocolitica*: molecular and evolutionary aspects. *J Bacteriol* 1999;181:6387–6395.
79. Kim H, Lee H, Shin D. The FeoA protein is necessary for the FeoB transporter to import ferrous iron. *Biochem Biophys Res Commun* 2012;423:733–738.
80. Iriarte M, Stainier I, Cornelis GR. The rpoS gene from *Yersinia enterocolitica* and its influence on expression of virulence factors. *Infect Immun* 1995;63:1840–1847.
81. Baranova N, Nikaido H. The baeSR two-component regulatory system activates transcription of the yegMNOB (mdtABCD) transporter gene cluster in *Escherichia coli* and increases its resistance to novobiocin and deoxycholate. *J Bacteriol* 2002;184:4168–4176.
82. Zhou D, Han Y, Qiu J, Qin L, Guo Z et al. Genome-wide transcriptional response of *Yersinia pestis* to stressful conditions simulating phagolysosomal environments. *Microbes Infect* 2006;8:2669–2678.
83. Chen S, Thompson KM, Francis MS. Environmental regulation of *Yersinia* pathophysiology. *Front Cell Infect Microbiol* 2016;6:25.
84. Kim TJ, Young BM, Young GM. Effect of flagellar mutations on *Yersinia enterocolitica* biofilm formation. *Appl Environ Microbiol* 2008;74:5466–5474.
85. Young GM, Smith MJ, Minnich SA, Miller VL. The *Yersinia enterocolitica* motility master regulatory operon, flhDC, is required for flagellin production, swimming motility, and swarming motility. *J Bacteriol* 1999;181:2823–2833.
86. Wuttge S, Bommer M, Jäger F, Martins BM, Jacob S et al. Determinants of substrate specificity and biochemical properties of the sn-glycerol-3-phosphate ATP binding cassette transporter (UgpB-AEC2) of *Escherichia coli*. *Mol Microbiol* 2012;86:908–920.